

Assignment 1 Machine Learning

Name: Justin Christian Kenan

Student ID: 2802399463

Class: LC01

Libraries

```
In [59]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder, StandardScaler, MinMaxScaler
```

Dataset

```
In [60]: dataset = pd.read_csv('covid_19_indonesia_time_series_all.csv')
dataset.head(10)
```

Out[60]:

	Date	Location ISO Code	Location	New Cases	New Deaths	New Recovered	New Active Cases	Total Cases	Total Deaths	Total Recovered
0	3/1/2020	ID-JK	DKI Jakarta	2	0	0	2	39	20	
1	3/2/2020	ID-JK	DKI Jakarta	2	0	0	2	41	20	
2	3/2/2020	IDN	Indonesia	2	0	0	2	2	0	
3	3/2/2020	ID-RI	Riau	1	0	0	1	1	0	
4	3/3/2020	ID-JK	DKI Jakarta	2	0	0	2	43	20	
5	3/3/2020	IDN	Indonesia	0	0	0	0	2	0	
6	3/3/2020	ID-JB	Jawa Barat	1	1	0	0	1	1	
7	3/3/2020	ID-RI	Riau	0	0	0	0	1	0	
8	3/4/2020	ID-JK	DKI Jakarta	2	0	0	2	45	20	
9	3/4/2020	IDN	Indonesia	0	0	0	0	2	0	

10 rows × 38 columns



Exploratory Data Analysis (EDA)

Dataset Structure

In [61]: *# Number of rows and columns in the dataset*
dataset.shape

Out[61]: (31822, 38)

In [62]: *# Data types of each column*
dataset.dtypes

```

Out[62]: Date                object
        Location ISO Code    object
        Location             object
        New Cases            int64
        New Deaths           int64
        New Recovered        int64
        New Active Cases     int64
        Total Cases           int64
        Total Deaths         int64
        Total Recovered      int64
        Total Active Cases   int64
        Location Level       object
        City or Regency      float64
        Province            object
        Country              object
        Continent            object
        Island               object
        Time Zone            object
        Special Status       object
        Total Regencies      int64
        Total Cities         float64
        Total Districts      int64
        Total Urban Villages float64
        Total Rural Villages float64
        Area (km2)           int64
        Population           int64
        Population Density    float64
        Longitude            float64
        Latitude             float64
        New Cases per Million float64
        Total Cases per Million float64
        New Deaths per Million float64
        Total Deaths per Million float64
        Total Deaths per 100rb float64
        Case Fatality Rate    object
        Case Recovered Rate   object
        Growth Factor of New Cases float64
        Growth Factor of New Deaths float64
        dtype: object

```

```

In [63]: # Descriptive statistical analysis on dataset
        dataset.describe()

```

Out[63]:

	New Cases	New Deaths	New Recovered	New Active Cases	Total Cases	Total Deat
count	31822.000000	31822.000000	31822.000000	31822.000000	3.182200e+04	31822.0000
mean	402.311388	9.920652	390.398498	1.992238	1.594500e+05	4564.7532
std	2320.629838	64.139080	2199.878802	1219.513355	6.264435e+05	17693.7313
min	0.000000	0.000000	0.000000	-29938.000000	1.000000e+00	0.0000
25%	3.000000	0.000000	2.000000	-12.000000	5.223250e+03	128.0000
50%	27.000000	0.000000	20.000000	0.000000	2.359650e+04	565.5000
75%	130.000000	3.000000	123.000000	19.000000	6.992775e+04	2189.0000
max	64718.000000	2069.000000	61361.000000	39165.000000	6.405044e+06	157876.0000

8 rows × 26 columns



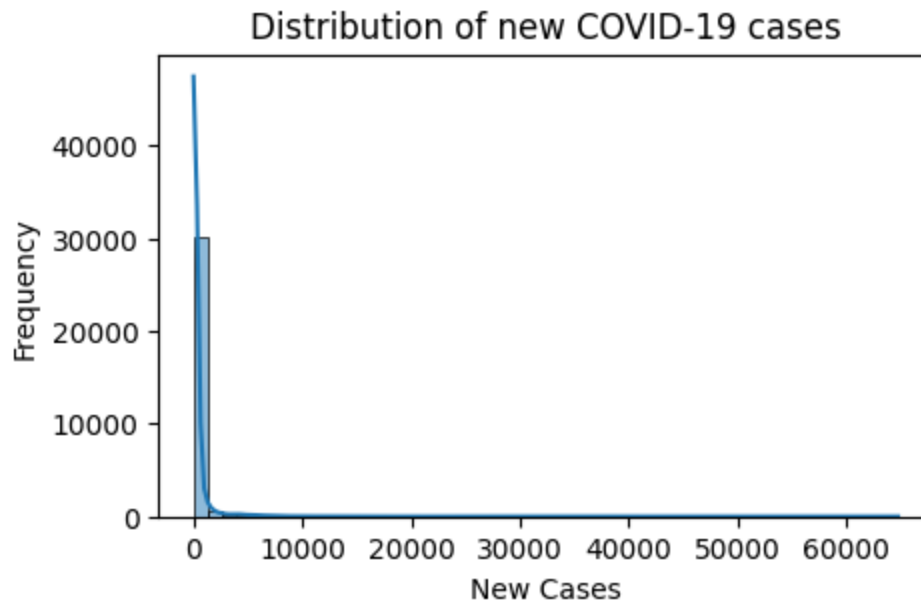
Checking Missing Values

In [64]: `dataset.isnull().sum().sort_values(ascending=False)`

```
Out[64]: City or Regency      31822
Special Status      27264
Growth Factor of New Deaths    3447
Growth Factor of New Cases    1939
Total Rural Villages      929
Island      929
Time Zone      929
Province      929
Total Urban Villages      904
Total Cities      901
Location ISO Code      0
Date      0
New Cases      0
Location      0
New Recovered      0
New Deaths      0
Total Deaths      0
Total Recovered      0
Total Active Cases      0
Location Level      0
New Active Cases      0
Total Cases      0
Total Districts      0
Total Regencies      0
Continent      0
Country      0
Population Density      0
Longitude      0
Area (km2)      0
Population      0
New Cases per Million      0
Latitude      0
Total Cases per Million      0
New Deaths per Million      0
Total Deaths per 100rb      0
Total Deaths per Million      0
Case Recovered Rate      0
Case Fatality Rate      0
dtype: int64
```

Histogram

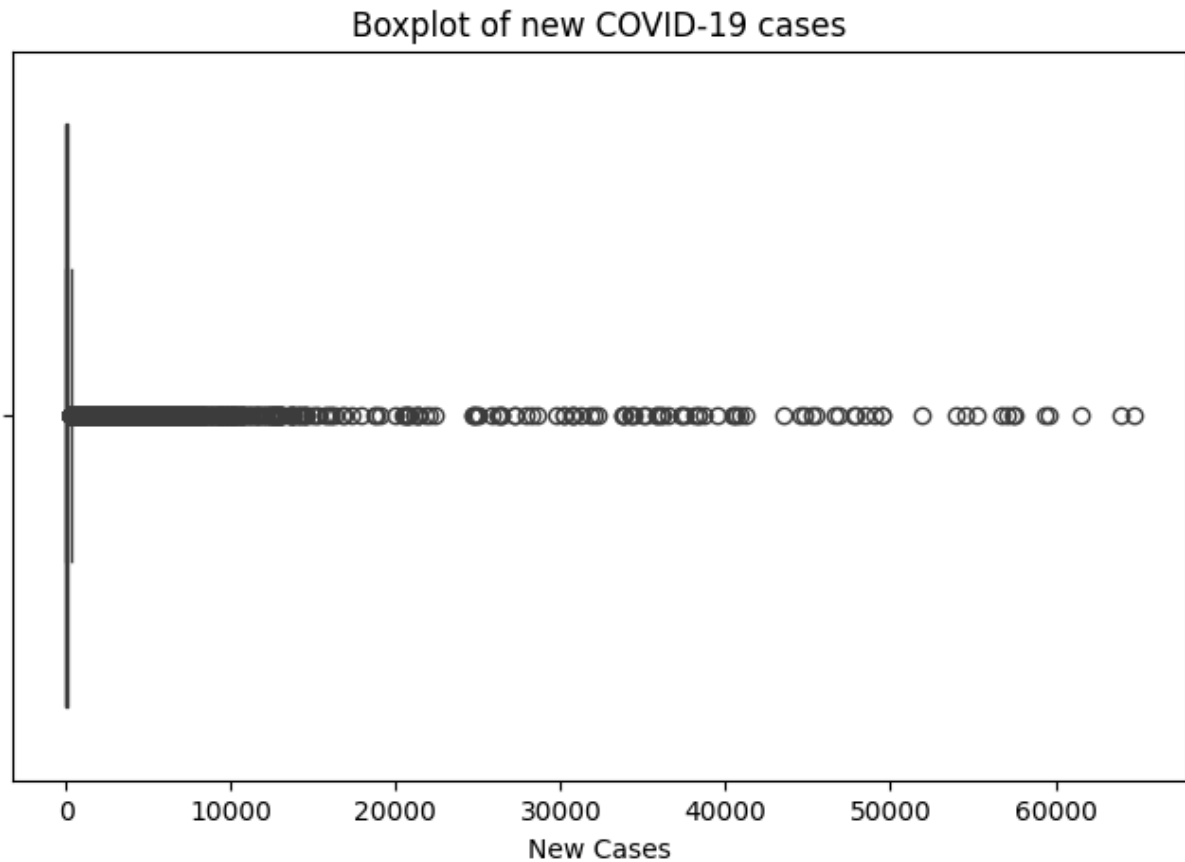
```
In [65]: plt.figure(figsize=(5,3))
plt.title("Distribution of new COVID-19 cases")
plt.xlabel("New Cases")
plt.ylabel("Frequency")
sns.histplot(dataset["New Cases"], bins=50, kde=True)
plt.show()
```



The distribution of new COVID-19 cases is showed skewed to the right. Most of the data has a low number of cases, but there are some very high spikes in cases.

Box Plot

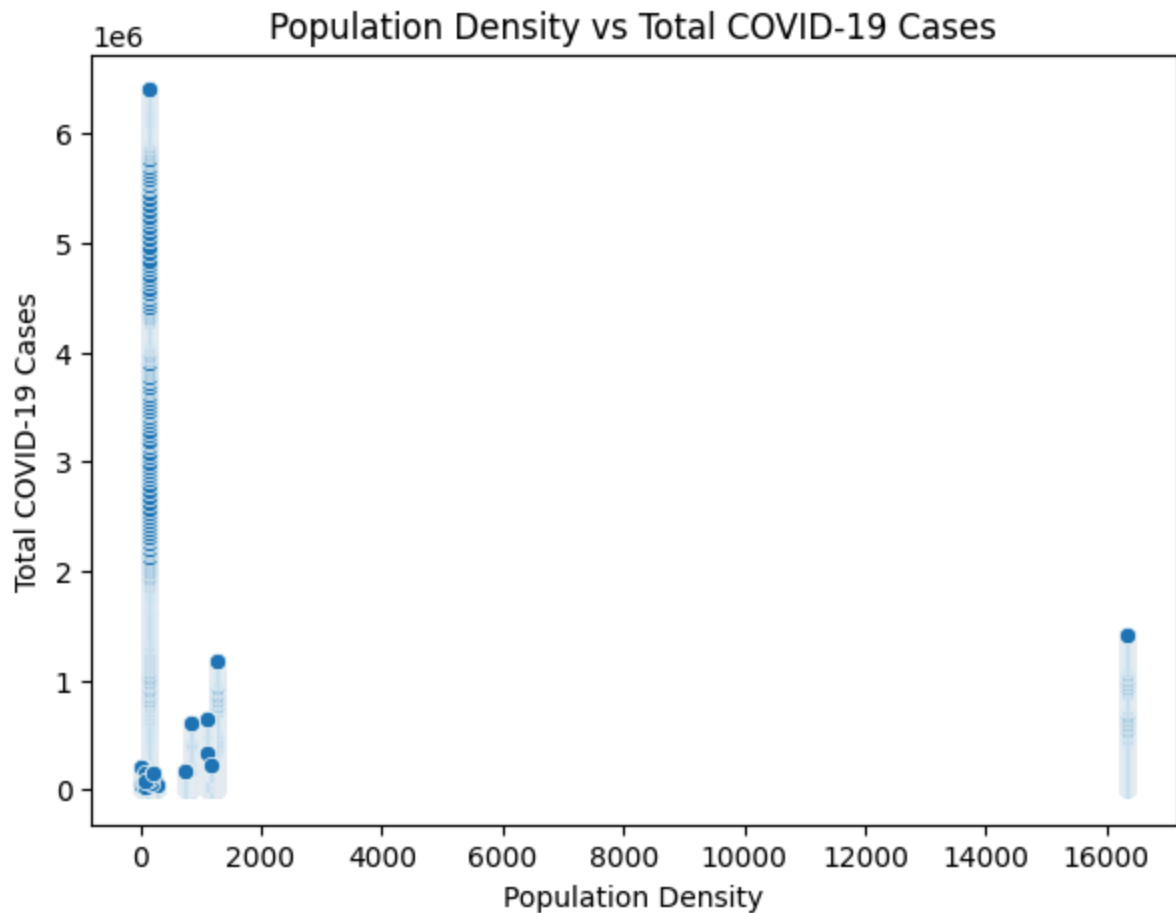
```
In [66]: plt.figure(figsize=(8, 5))
plt.title("Boxplot of new COVID-19 cases")
sns.boxplot(x=dataset["New Cases"])
plt.show()
```



The boxplot shows a large number of outliers in the new COVID-19 cases feature. This occurs because the dataset contains several days with very high spikes in cases compared to the rest of the data.

Scatter Plot

```
In [67]: plt.figure(figsize=(7, 5))
plt.title("Population Density vs Total COVID-19 Cases")
plt.xlabel("Population Density")
plt.ylabel("Total COVID-19 Cases")
sns.scatterplot(x="Population Density", y="Total Cases", data=dataset)
plt.show()
```



Scatter Plot shows many observations share the same population density values with the result vertical cluster of points, because the dataset records multiple times entries for the same provinces with population density remains constant while the total number of COVID-19 cases changes over time.

Interesting Insight

1. There several significant COVID-19 cases during the pandemic, so the Distribution of new cases shows an abnormal pattern or skewed.
2. There's a positive correlation between Population density and Total cases, it's mean shows that areas with higher population density have a greater potential for the spread of the COVID-19 virus.
3. The increase in the number or COVID-19 cases followed by an increase in the number of the deaths.

Data Preprocessing

Handling missing values

The strategy for filling in missing data in the dataset is using mean imputation to maintain data distribution without deleting too many rows of data.

```
In [68]: dataset = dataset.fillna(dataset.mean(numeric_only=True))
```

Encoding categorical features using Label Encoder

Label Encoding used to transform categorical features into numerical values with assigns a unique integer into each category in a feature, also this method simple and efficient which that suitable for categorical variables with unique categories (provinces, island, location level).

```
In [69]: label_encoder = LabelEncoder()

dataset['Province'] = label_encoder.fit_transform(dataset['Province'].astype(str))
dataset['Island'] = label_encoder.fit_transform(dataset['Island'].astype(str))
dataset['Location Level'] = label_encoder.fit_transform(dataset['Location Level'].a
```

Normalize (feature scalling)

StandardScaler transforms the data, which has a standard deviation of 1 and mean of 0. The numerical features in the dataset has different ranges so that normalize will improve the performance of machine learning algorithms that sensitive to feature scalling.

```
In [70]: scaler = StandardScaler()
numeric_cols = ['Population', 'Population Density', 'New Cases', 'Total Cases']
dataset[numeric_cols] = scaler.fit_transform(dataset[numeric_cols])
```

Result

```
In [ ]: dataset.head(15)
```

Out[]:

	Date	Location ISO Code	Location	New Cases	New Deaths	New Recovered	New Active Cases	Total Cases	Total Deaths
0	3/1/2020	ID-JK	DKI Jakarta	-0.172504	0	0	2	-0.254474	20
1	3/2/2020	ID-JK	DKI Jakarta	-0.172504	0	0	2	-0.254471	20
2	3/2/2020	IDN	Indonesia	-0.172504	0	0	2	-0.254533	0
3	3/2/2020	ID-RI	Riau	-0.172935	0	0	1	-0.254535	0
4	3/3/2020	ID-JK	DKI Jakarta	-0.172504	0	0	2	-0.254468	20
5	3/3/2020	IDN	Indonesia	-0.173366	0	0	0	-0.254533	0
6	3/3/2020	ID-JB	Jawa Barat	-0.172935	1	0	0	-0.254535	1
7	3/3/2020	ID-RI	Riau	-0.173366	0	0	0	-0.254535	0
8	3/4/2020	ID-JK	DKI Jakarta	-0.172504	0	0	2	-0.254464	20
9	3/4/2020	IDN	Indonesia	-0.173366	0	0	0	-0.254533	0
10	3/4/2020	ID-JB	Jawa Barat	-0.172935	0	0	1	-0.254533	1
11	3/4/2020	ID-RI	Riau	-0.173366	0	0	0	-0.254535	0
12	3/5/2020	ID-JK	DKI Jakarta	-0.173366	1	0	-1	-0.254464	21
13	3/5/2020	IDN	Indonesia	-0.173366	0	0	0	-0.254533	0
14	3/5/2020	ID-JB	Jawa Barat	-0.172935	0	0	1	-0.254531	1

15 rows × 38 columns

